ICT Corner

CHEMICAL KINETICS

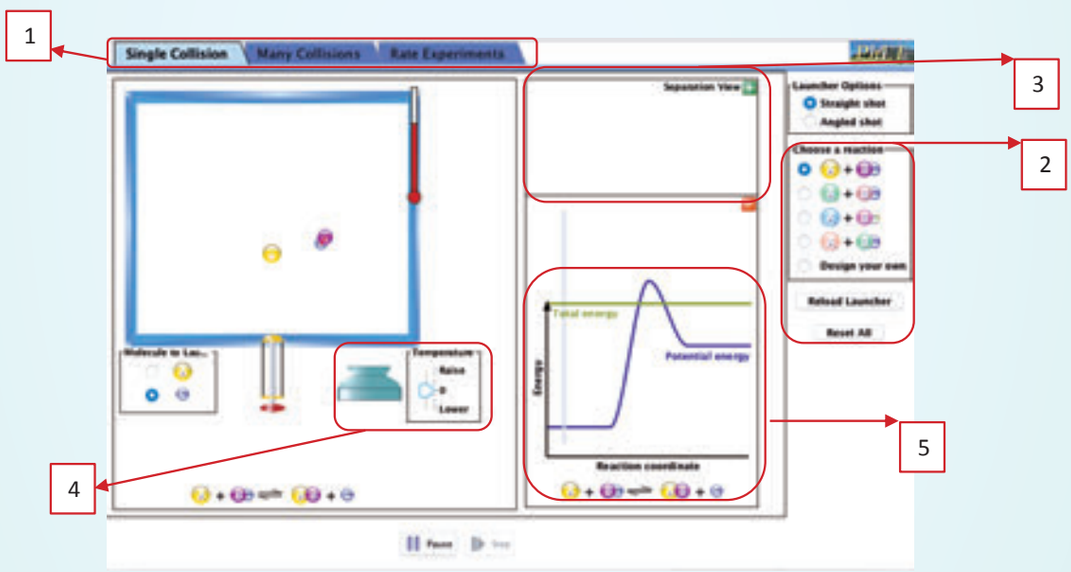
By using this tool you can understand collision and the activation energy. You can also perform virtual kinetic experiment to understand the effect of temperature on reaction rate.

Please go to the URL
https://phet.colorado.edu/en/simulation/legacy/reactions-and-rates
(or) Scan the QR code on the right side

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Steps

- Open the Browser and type the URL given (or) Scan the QR Code. You can see a webpage which displays the java applet called reactions and rates. You can click it and you will get a window as shown in the figure. This applet contains three modules which can be selected by clicking the appropriate tab (box1).
- Select single collision tab (tab 1) to visualise collision between two molecules. You can visualise the progress of the reaction (box 5) by varying the temperature using the slider (box 2). You can visualise that the raise in temperature, will raise the energy of the system and allows the reactants to cross the energy barrier to form products. You can repeat this simulation with more molecules in the many collisions tab (box 1).
- You can also perform virtual kinetic experiment, using rate experiments mode. Choose the types reacting molecules and their stoichiometry using the options provided (box 2). The number of reactant and product molecules at a given time will be displayed in panel (box-3). You can see the effect of temperature on reaction rate by varying the temperature (box 4).



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